

Permutation Inversions

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to count the number of permutations of $1, 2, \dots, n$ that have exactly k inversions (i.e., pairs of elements in the wrong order).

For example, when $n = 4$ and $k = 3$, there are 6 such permutations:

- $[1, 4, 3, 2]$
- $[2, 3, 4, 1]$
- $[2, 4, 1, 3]$
- $[3, 1, 4, 2]$
- $[3, 2, 1, 4]$
- $[4, 1, 2, 3]$

Input

The only input line has two integers n and k .

Output

Print the answer modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 500$
- $0 \leq k \leq \frac{n(n-1)}{2}$

Example

Input	Output
4 3	6